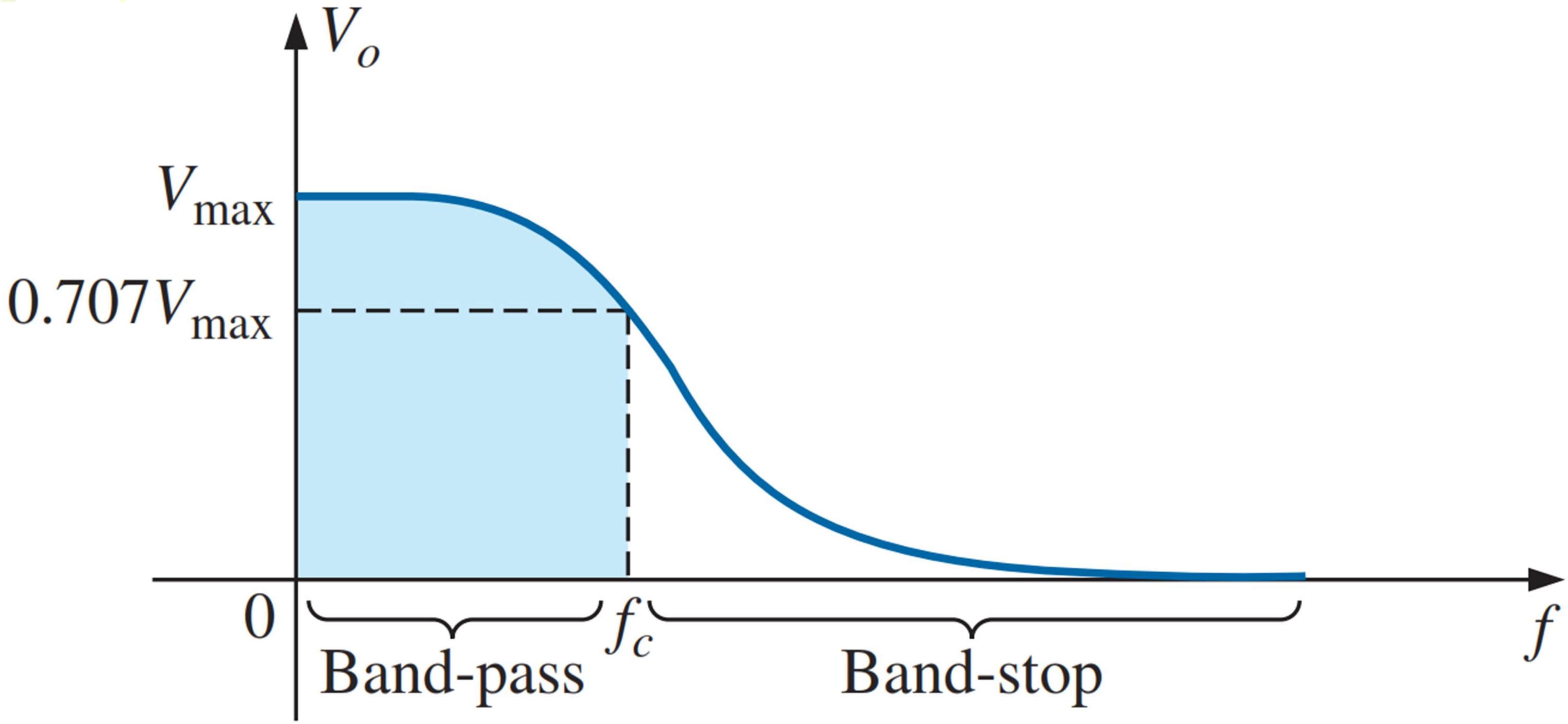
Filters

Passive Filters

- A filter is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.
- Passive filter consists of only passive element R, L and C.
- As a frequency-selective device, a filter can be used to limit the frequency spectrum of a signal to some specified band of frequencies.
- Filters are the circuits used in radio and TV receivers to allow us to select one desired signal out of a multitude of broadcast signals in the environment

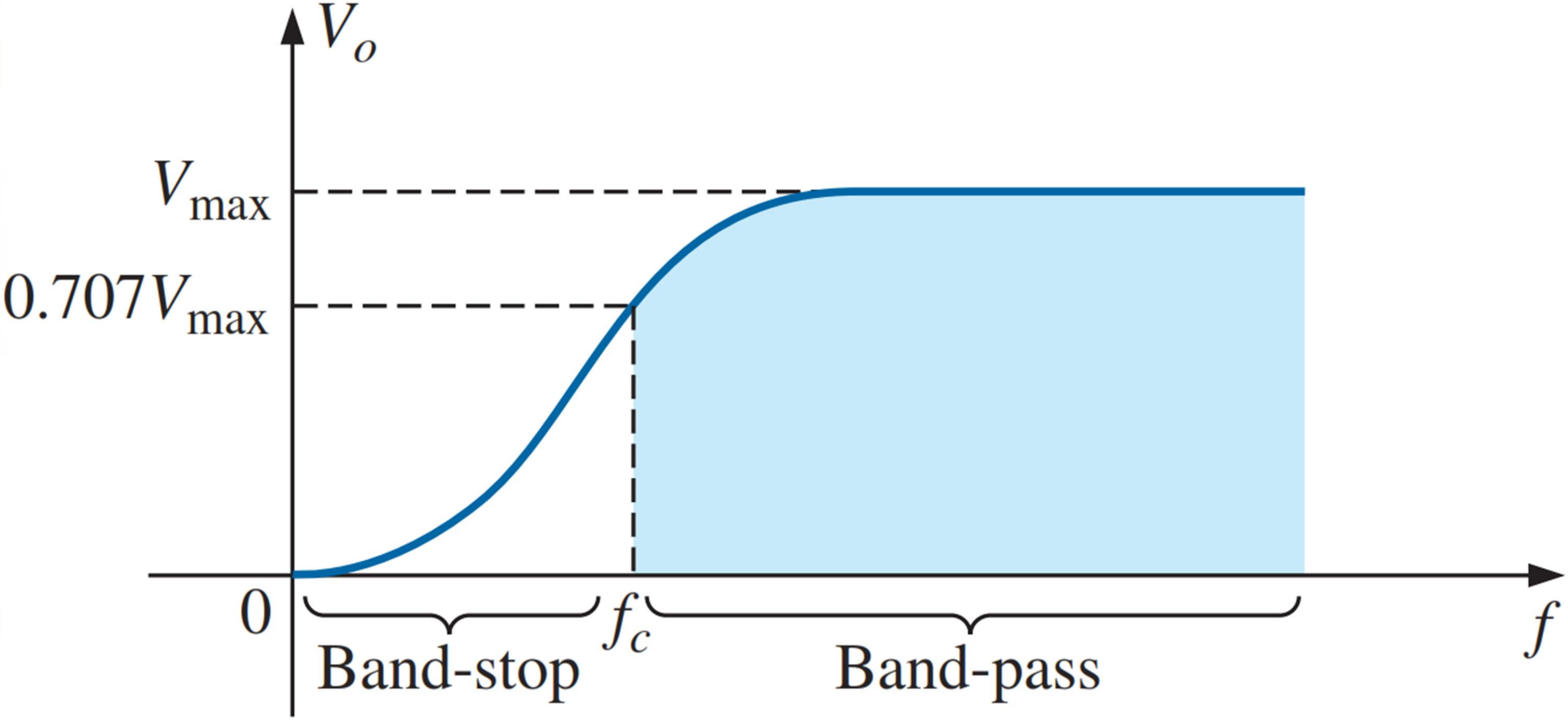
Low-pass filter

1. A *low-pass filter* passes low frequencies and stops high frequencies,



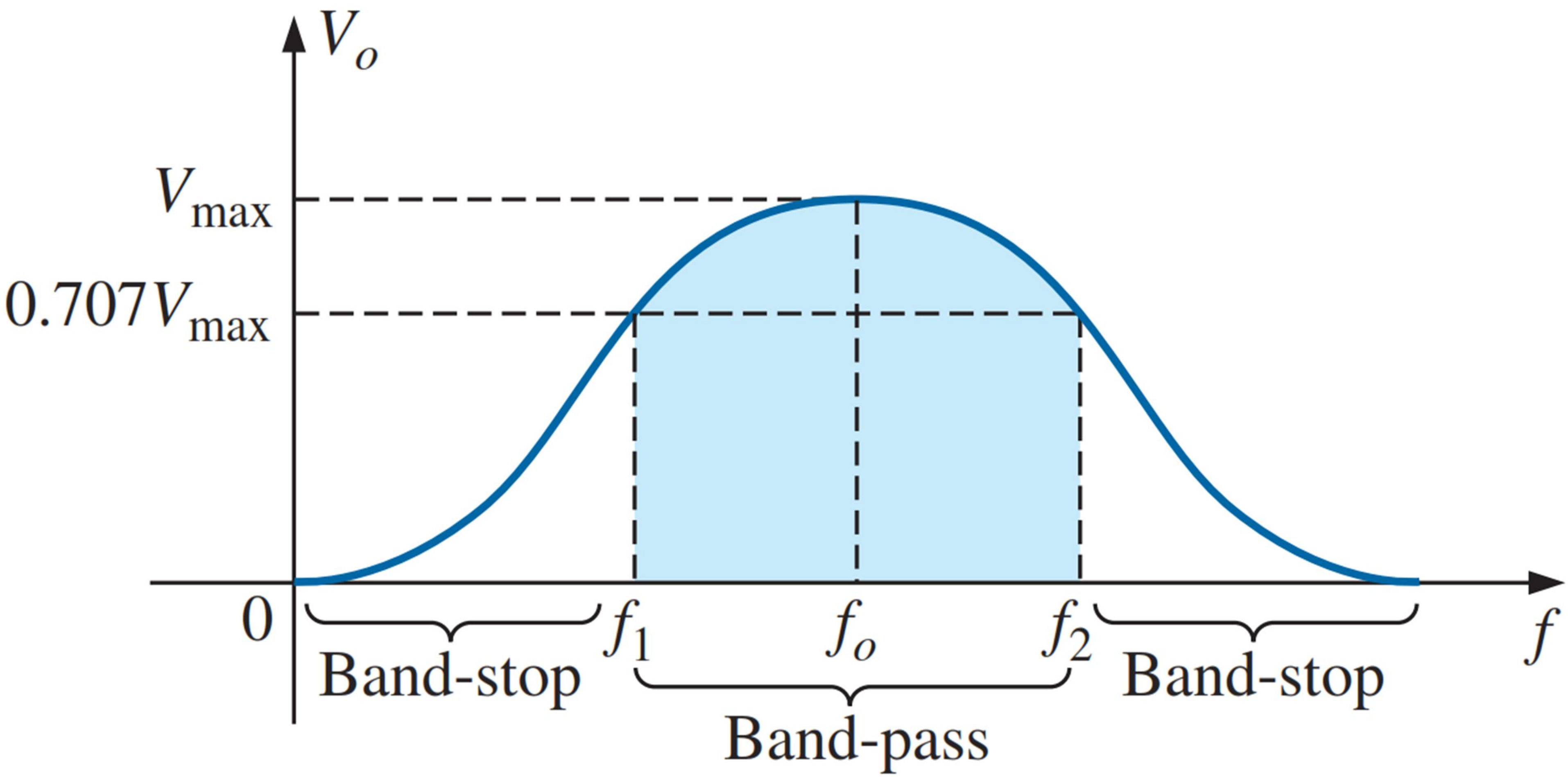
High-pass filter:

2. A *high-pass filter* passes high frequencies and rejects low frequencies,



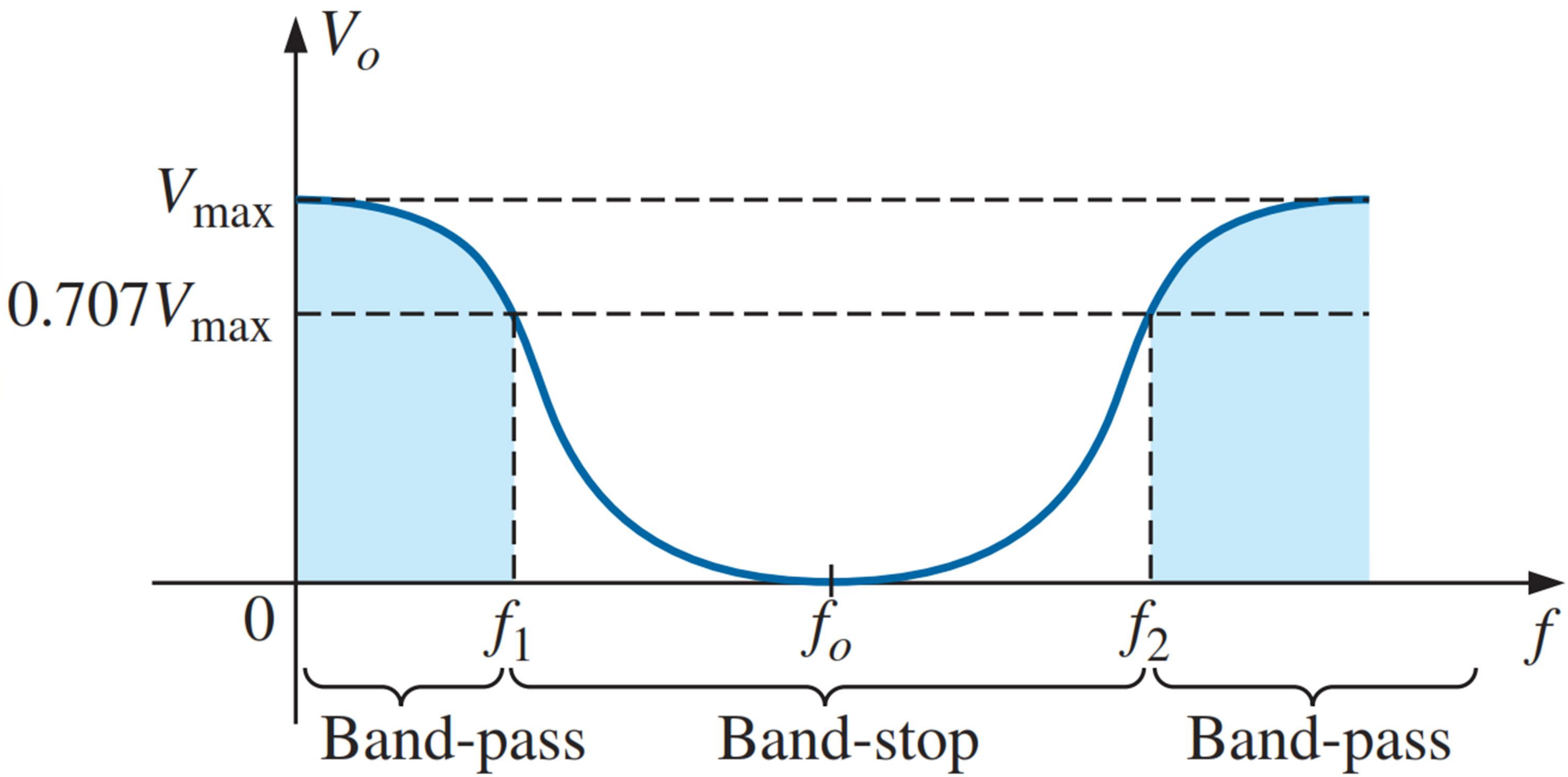
Band-pass filter

3. A *band-pass filter* passes frequencies within a frequenc y band and blocks or attenuates frequencies outside the band



Band-stop filter

4. A *band-stop filter* passes frequencies outside a frequency band and blocks or attenuates frequencies within the band



R-C LOW-PASS FILTER

The *R-C* filter, incredibly simple in design, can be used as a low-pass or a high-pass filter. If the output is taken off the capacitor, as shown in Fig. 22.10, it responds as a low-pass filter. If the positions of the resistor and capacitor are interchanged and the output is taken off the resistor, the response is that of a high-pass filter.

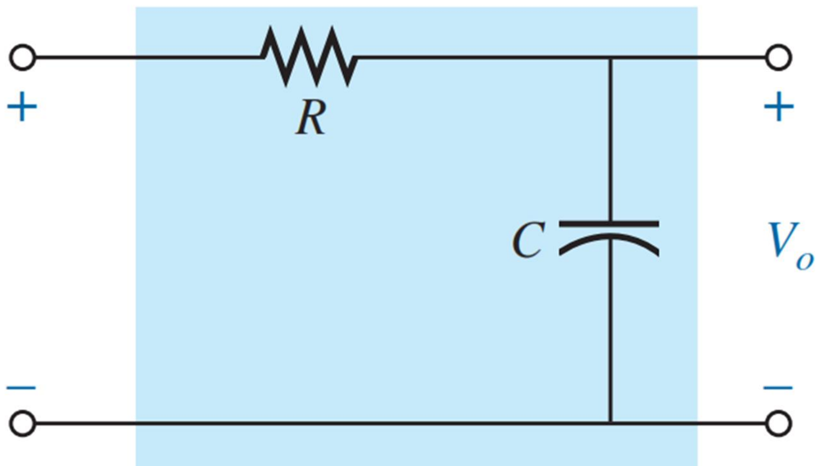


FIG. 22.10
Low-pass filter.

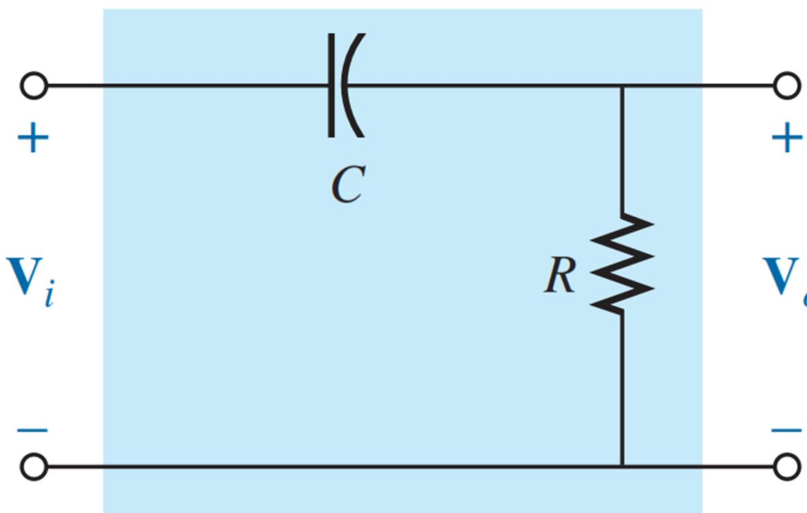


FIG. 22.21
High-pass filter.

At $f = 0$ Hz,

$$X_C = \frac{1}{2\pi fC} = \infty\,\Omega$$

and the open-circuit equivalent can be substituted for the capacitor, as shown in Fig. 22.11, resulting in $\mathbf{V}_o = \mathbf{V}_i$.

At very high frequencies, the reactance is

$$X_C = \frac{1}{2\pi fC} \cong 0\,\Omega$$

and the short-circuit equivalent can be substituted for the capacitor, as shown in Fig. 22.12, resulting in $\mathbf{V}_o = 0$ V.

A plot of the magnitude of V_o versus frequency results in the curve in Fig. 22.13. Our next goal is now clearly defined: Find the frequency at which the transition takes place from a pass-band to a stop-band.

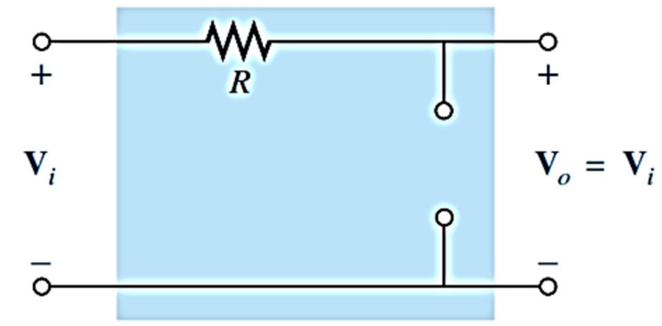


FIG. 22.11
R-C low-pass filter at low frequencies.

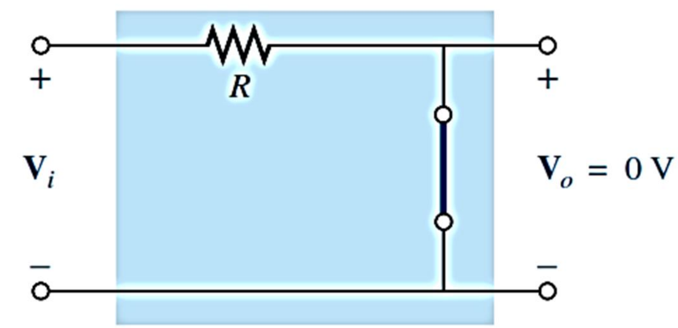


FIG. 22.12
R-C low-pass filter at high frequencies.

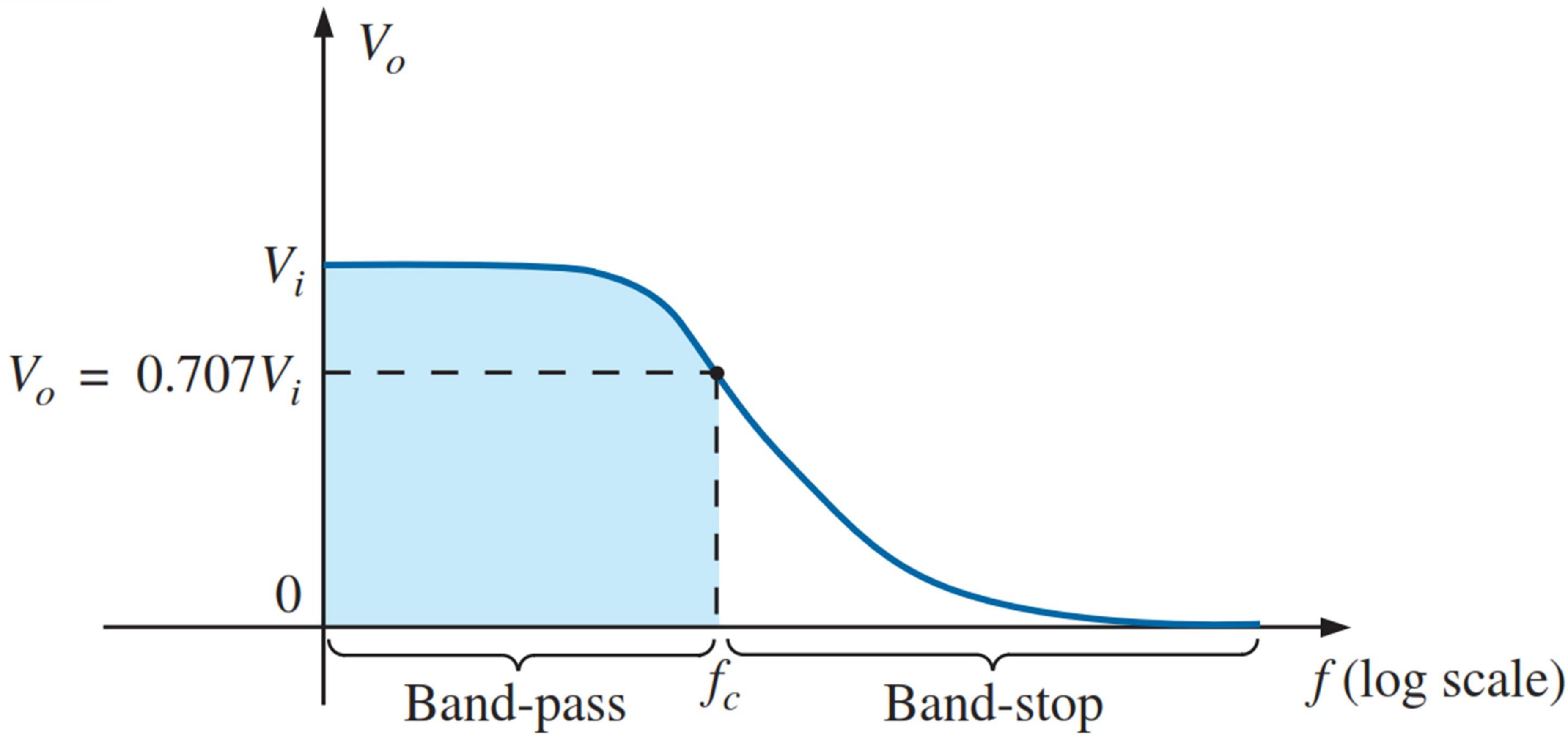


FIG. 22.13

V_o versus frequency for a low-pass R-C filter.

A normalized plot in the filter domain can be obtained by dividing the plotted quantity such as V_o in Fig. 22.13 with the applied voltage V_i for the frequency range of interest. Since the maximum value of V_o for the low-pass filter in Fig. 22.10 is V_i, each level of V_o in Fig. 22.11 is divided by the level of V_i. The result is the plot of $A_v = V_o/V_i$ in Fig. 22.14. Note that the maximum value is 1 and the cutoff frequency is defined at the 0.707 level.

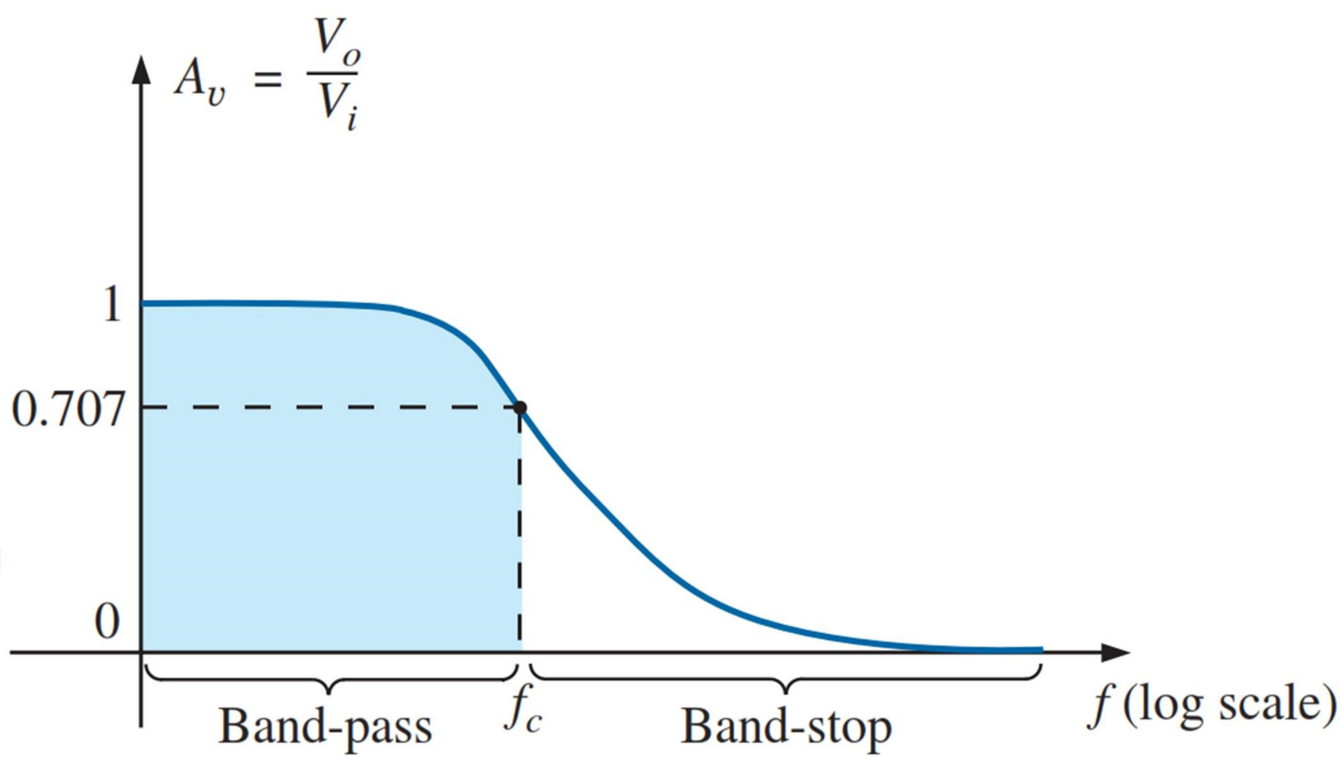


FIG. 22.14

Normalized plot of Fig. 22.13.

The magnitude of the ratio V_o/V_i is therefore determined by

$$\boxed{A_v = \frac{V_o}{V_i} = \frac{X_C}{\sqrt{R^2 + X_C^2}} = \frac{1}{\sqrt{\left(\frac{R}{X_C}\right)^2 + 1}}}$$

and the phase angle θ by

$$\boxed{\theta = \tan^{-1}\frac{X_C}{R}}$$

For the special frequency at which $X_C = R$, the magnitude becomes

$$A_v = \frac{V_o}{V_i} = \frac{1}{\sqrt{\left(\frac{R}{X_C}\right)^2 + 1}} = \frac{1}{\sqrt{1 + 1}} = \frac{1}{\sqrt{2}} = 0.707$$

which defines the critical or cutoff frequency in Fig. 22.14.

The frequency at which $X_C = R$ is determined by

$$\frac{1}{2\pi f_c C} = R$$

and

$$\boxed{f_c = \frac{1}{2\pi RC}} \qquad \textbf{(22.16)}$$

R-C High Pass Filter

The *R-C* filter, incredibly simple in design, can be used as a low-pass or a high-pass filter. If the output is taken off the capacitor, as shown in Fig. 22.10, it responds as a low-pass filter. If the positions of the resistor and capacitor are interchanged and the output is taken off the resistor, the response is that of a high-pass filter.

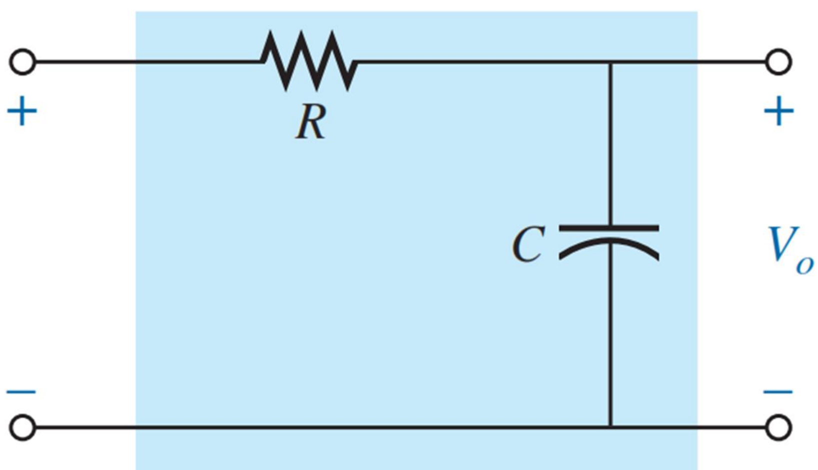


FIG. 22.10
Low-pass filter.

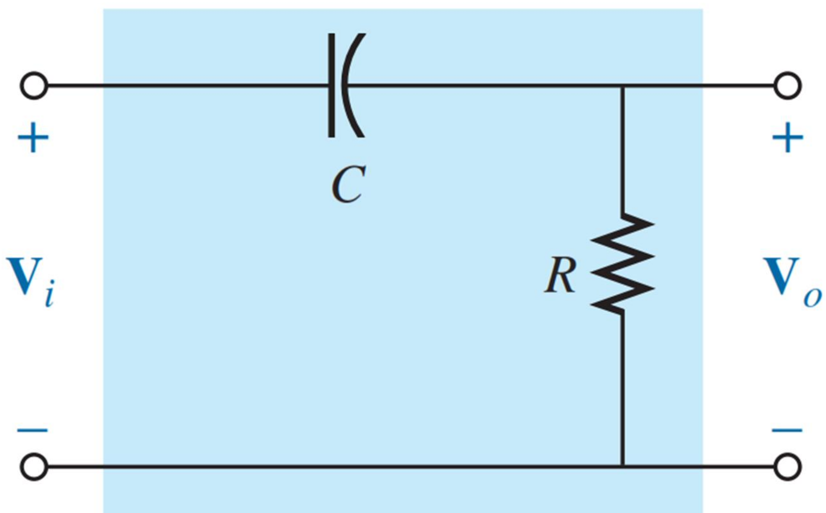


FIG. 22.21
High-pass filter.

At very high frequencies, the reactance of the capacitor is very small, and the short-circuit equivalent can be substituted, as shown in Fig. 22.22. The result is that $\mathbf{V}_o = \mathbf{V}_i$.

At $f = 0$ Hz, the reactance of the capacitor is quite high, and the open-circuit equivalent can be substituted, as shown in Fig. 22.23. In this case, $\mathbf{V}_o = 0$ V.

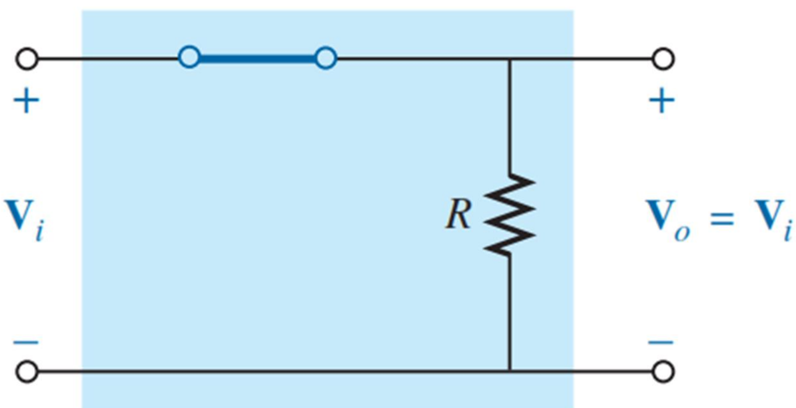


FIG. 22.22

R-C high-pass filter at very high frequencies.

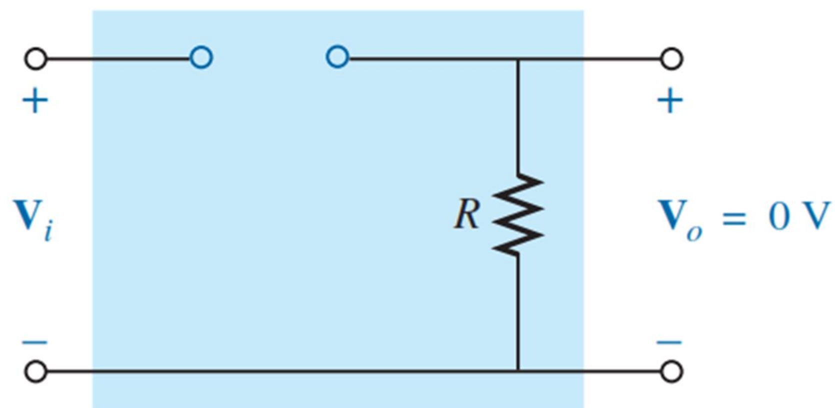


FIG. 22.23

R-C high-pass filter at $f = 0$ Hz.

A plot of the magnitude versus frequency is provided in Fig. 22.24, with the normalized plot in Fig. 22.25.

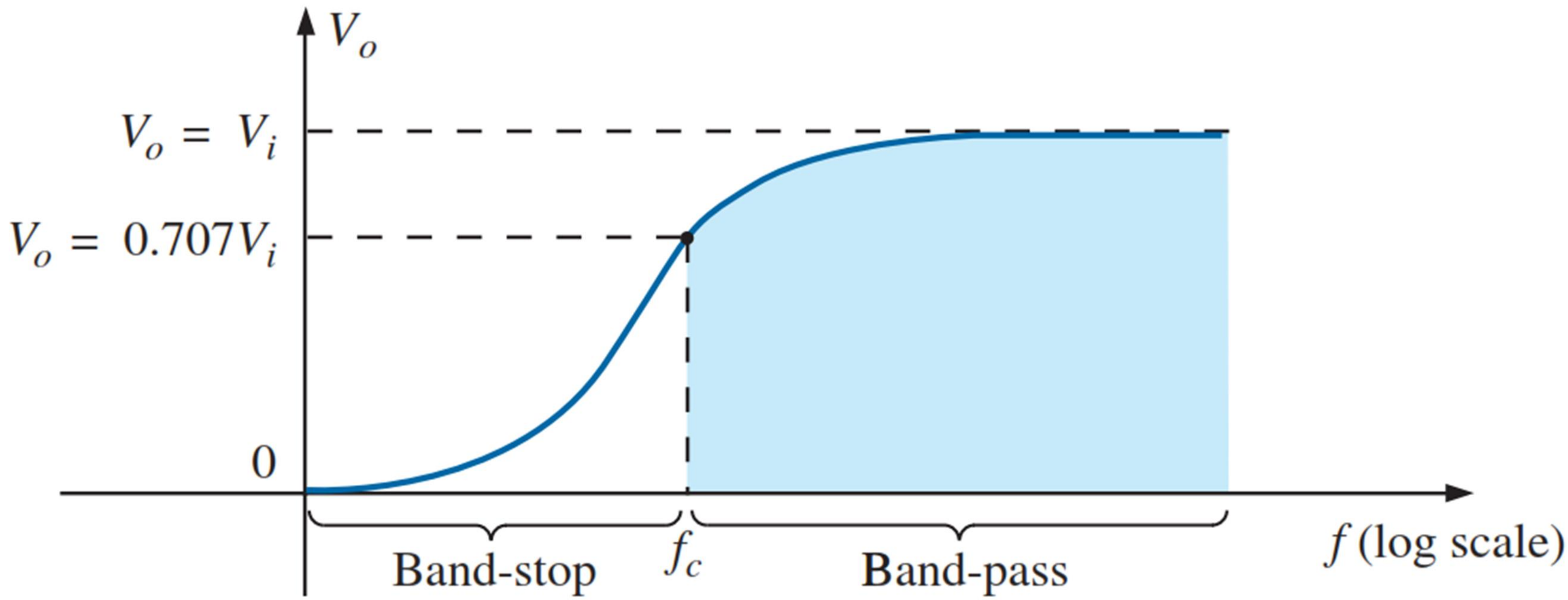


FIG. 22.24

V_o versus frequency for a high-pass R-C filter.

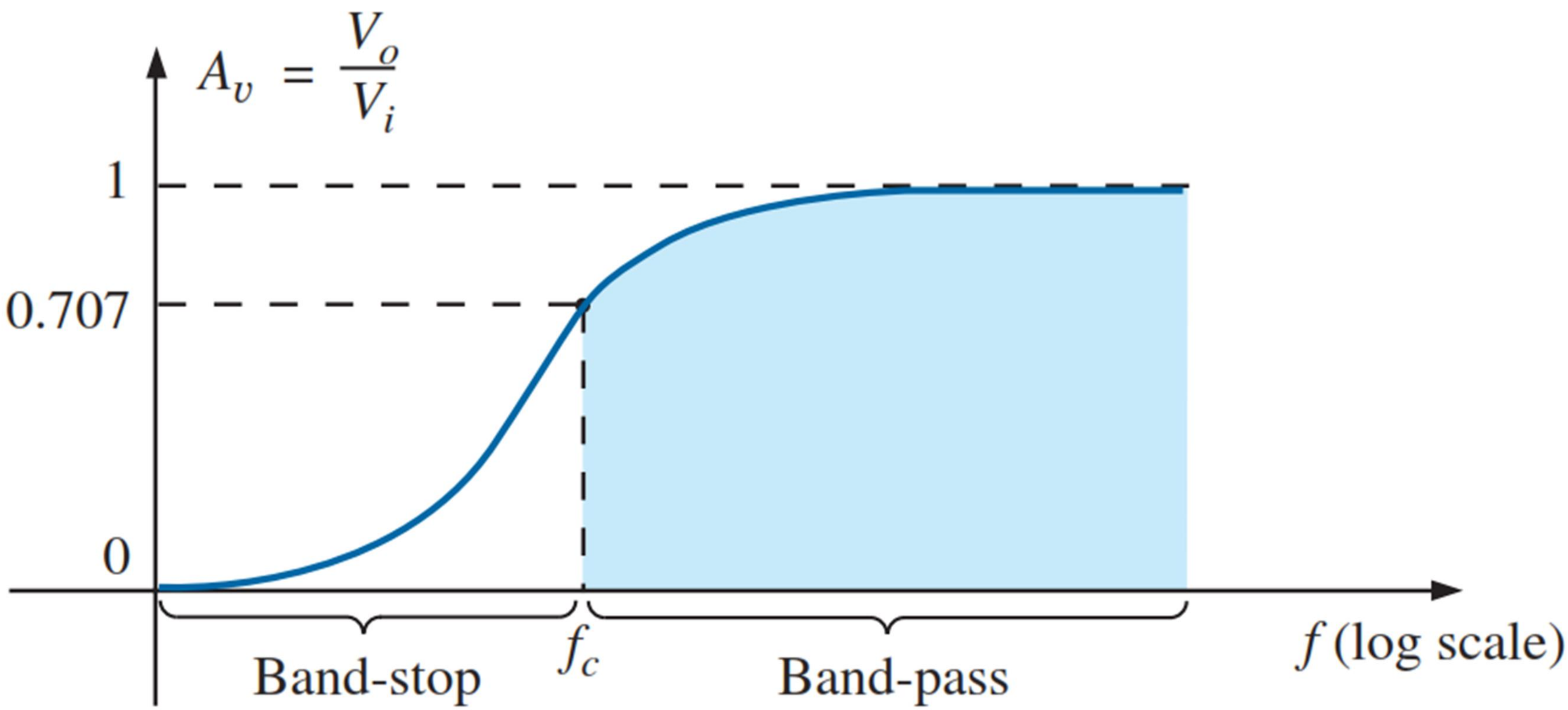


FIG. 22.25

Normalized plot of Fig. 22.24.

The magnitude of the ratio $\mathbf{V}_o/\mathbf{V}_i$ is therefore determined by

$$\boxed{A_v = \frac{V_o}{V_i} = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{1}{\sqrt{1 + \left(\frac{X_C}{R}\right)^2}}}$$

and the phase angle θ by

$$\boxed{\theta = \tan^{-1}\frac{X_C}{R}}$$

The frequency at which $X_C = R$ is determined by

$$X_C = \frac{1}{2\pi f_c C} = R$$

and

$$\boxed{f_c = \frac{1}{2\pi RC}} \tag{22.20}$$

For the high-pass R-C filter, the application of any frequency greater than f_c results in an output voltage V_o that is at least 70.7% of the magnitude of the input signal. For any frequency below f_c, the output is less than 70.7% of the applied signal.

EXAMPLE 22.6 Given $R = 20\text{ k}\Omega$ and $C = 1200\text{ pF}$:

a. Sketch the normalized plot if the filter is used as both a high-pass and a low-pass filter.
b. Sketch the phase plot for both filters in part (a).
c. Determine the magnitude and phase of $\mathbf{A}_v = \mathbf{V}_o/\mathbf{V}_i$ at $f = \frac{1}{2}f_c$ for the high-pass filter.

Solutions:

a. $$f_c = \frac{1}{2\pi RC} = \frac{1}{(2\pi)(20\text{ k}\Omega)(1200\text{ pF})} = \mathbf{6631.46\ Hz}$$

The normalized plots appear in Fig. 22.28.

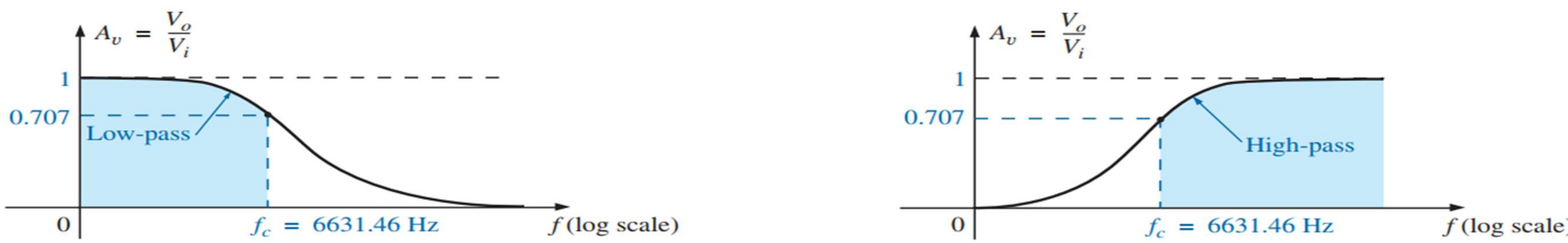


FIG. 22.29
Phase plots for a low-pass and a high-pass filter using the same elements.

c. $$f = \frac{1}{2}f_c = \frac{1}{2}(6631.46\text{ Hz}) = 3315.73\text{ Hz}$$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{(2\pi)(3315.73\text{ Hz})(1200\text{ pF})} \cong 40\text{ k}\Omega$$

$$A_v = \frac{V_o}{V_i} = \frac{1}{\sqrt{1+\left(\frac{X_C}{R}\right)^2}} = \frac{1}{\sqrt{1+\left(\frac{40\text{ k}\Omega}{20\text{ k}\Omega}\right)^2}} = \frac{1}{\sqrt{1+(2)^2}} = \frac{1}{\sqrt{5}} = 0.4472$$

$$\theta = \tan^{-1}\frac{X_C}{R} = \tan^{-1}\frac{40\text{ k}\Omega}{20\text{ k}\Omega} = \tan^{-1}2 = 63.43^\circ$$

and $$\mathbf{A}_v = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \mathbf{0.447\angle 63.43^\circ}$$

Band-Pass Filters

It is a filter that allows a certain band of frequencies to pass through and attenuates all other frequencies below and above the passband. This passband is known as the bandwidth of the filter. As seen, it is obtained by cascading a high-pass RC filter to a low-pass RC filter. It is shown in Fig. 22.35 along with its response curve. The passband of this filter is given by the band of frequencies lying between f_{c1} and f_{c2}. Their values are given by $f_{c1}=1/2\pi R_1C_1$ and $f_{c2}=1/2\pi R_2C_2$

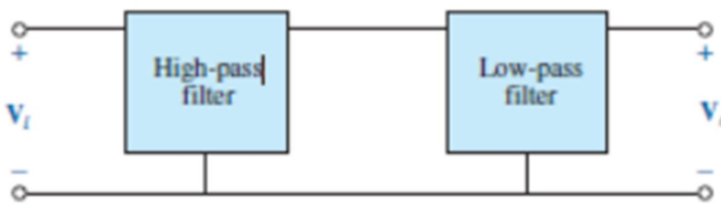
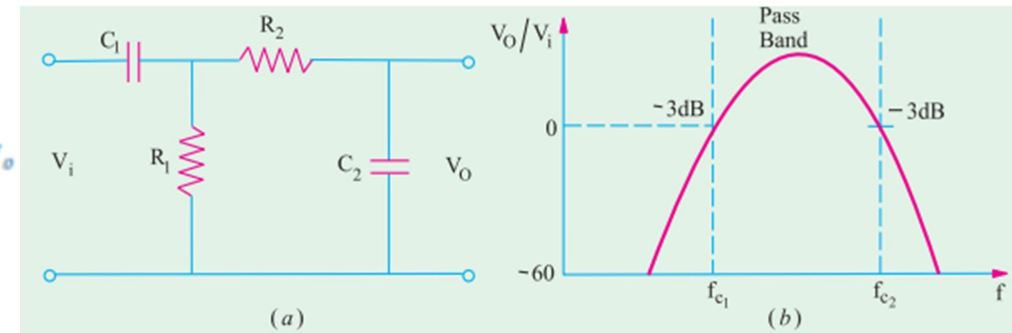
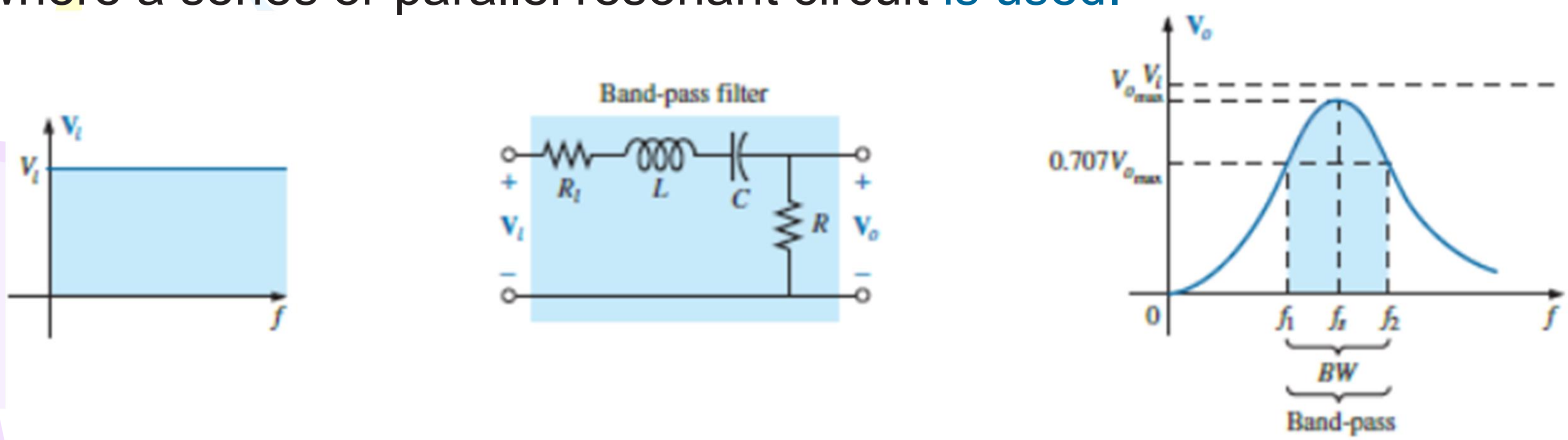


FIG. 22.35
Band-pass filter.



The most direct way to obtain the pass-band characteristics is shown in Fig. 22.30, where a series or parallel resonant circuit is used.



$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

$$Q_s = \frac{X_L}{R + R_l}$$

$$BW = \frac{f_s}{Q_s}$$

FIG. 22.30
Series resonant band-pass filter.

As a first approximation that is acceptable for most practical applications, it can be assumed that the resonant frequency bisects the bandwidth.

EXAMPLE 22.7

a. Determine the frequency response for the voltage V_o for the series circuit in Fig. 22.32.

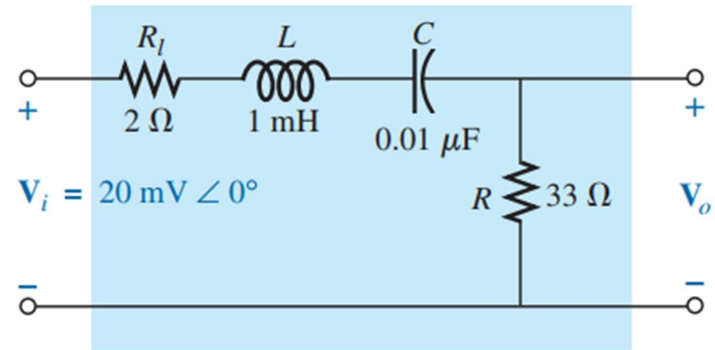


FIG. 22.32
Series resonant band-pass filter for Example 22.7.

Solutions:

a. $$f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(1\ \text{mH})(0.01\ \mu\text{F})}} = \mathbf{50{,}329.21\ Hz}$$

$$Q_s = \frac{X_L}{R + R_l} = \frac{2\pi(50{,}329.21\ \text{Hz})(1\ \text{mH})}{33\ \Omega + 2\ \Omega} = \mathbf{9.04}$$

$$BW = \frac{f_s}{Q_s} = \frac{50{,}329.21\ \text{Hz}}{9.04} = \mathbf{5.57\ kHz}$$

At resonance:

$$V_{o_{\max}} = \frac{RV_i}{R + R_l} = \frac{33\ \Omega(V_i)}{33\ \Omega + 2\ \Omega} = 0.943V_i = 0.943(20\ \text{mV})$$
$$= \mathbf{18.86\ mV}$$

At the cutoff frequencies:

$$V_o = (0.707)(0.943V_i) = 0.667V_i = 0.667(20\ \text{mV})$$
$$= \mathbf{13.34\ mV}$$

Note Fig. 22.33.

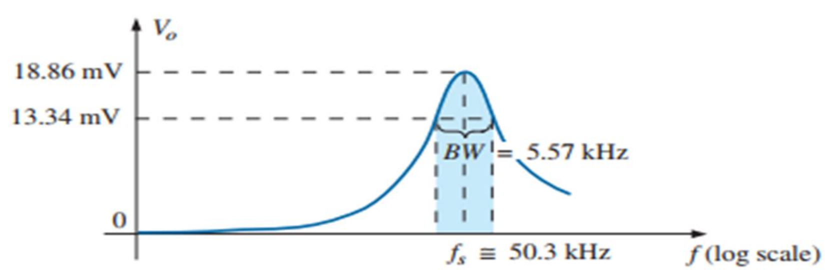
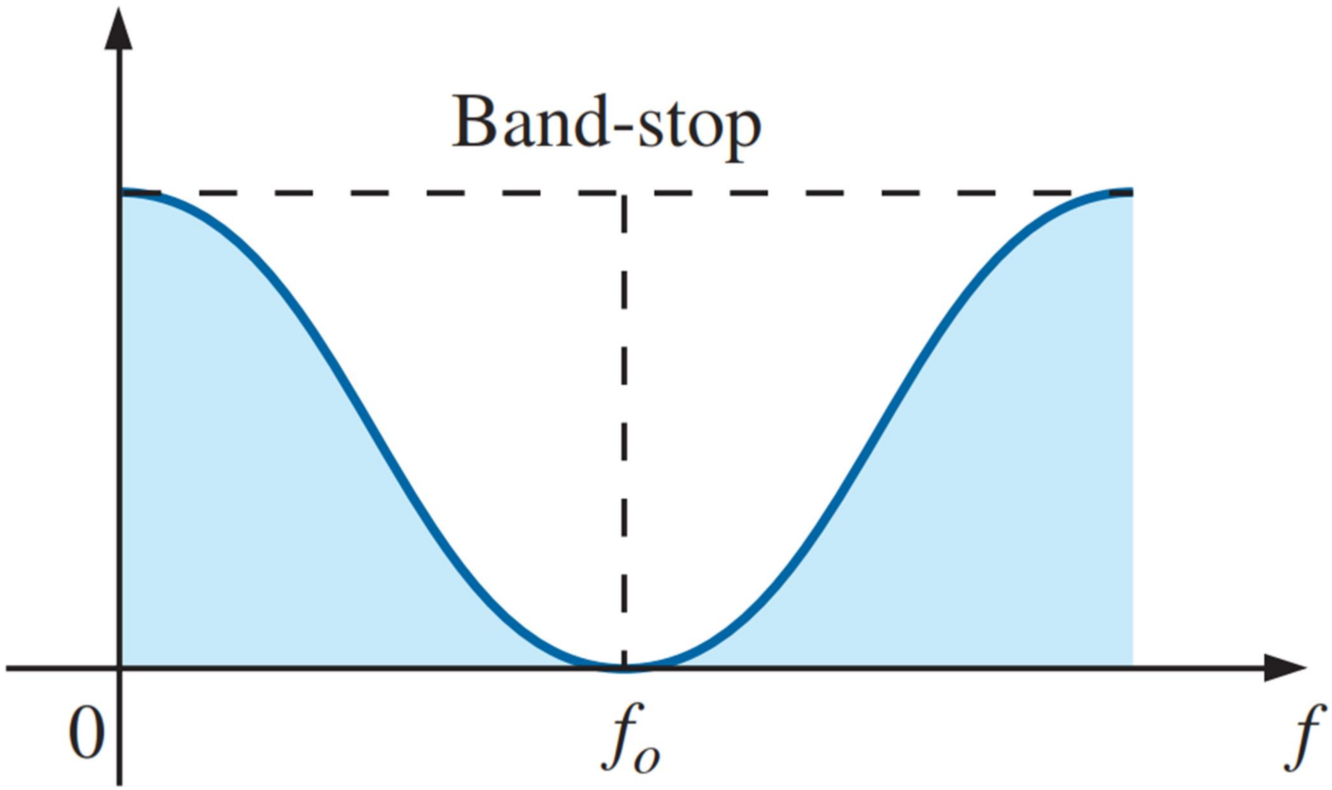


FIG. 22.33
Band-pass response for the network.

Band-stop Filters



Band-stop characteristics.